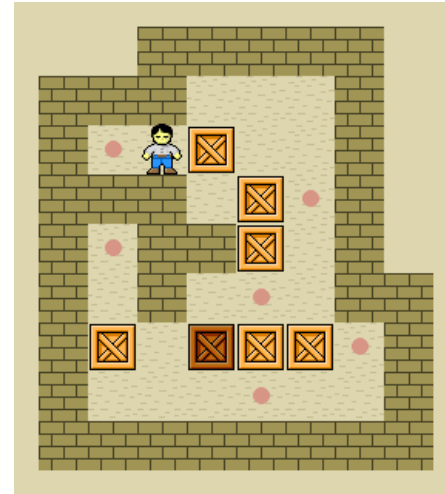




Introduction to the Course

CS 175: Projects in AI

Kalev Kask

Based on slides by Profs. Sameer Singh, Roy Fox

Projects in AI

Course Information

Logistics

- ❑ Course webpage: <https://canvas.eee.uci.edu/courses/82870>
- ❑ First 3-4 weeks, lectures on academic background material
 - ❑ Reinforcement Learning
 - ❑ Deep RL
 - ❑ Other topics, e.g. heuristics, search, etc.
- ❑ Discussion section to help with practical issues
- ❑ Project status/progress meetings weeks 5-10
- ❑ Two assignments
 - ❑ Testing mastery of basics
- ❑ First couple of weeks – lots of reading, thinking

Assignments

- ❑ Week 3
 - ❑ Team formation
 - ❑ Assignment 1
- ❑ Week 4
 - ❑ Project proposal
- ❑ Week 5
 - ❑ Assignment 2
- ❑ Week 8
 - ❑ Progress Report
- ❑ Week 10
 - ❑ Short Project Presentation
- ❑ Week 11
 - ❑ Final Project Report

Timeline

- ❑ Weeks 1-3

- ❑ Set up environment, try out simple code
- ❑ Form teams
- ❑ Decide on topic

- ❑ Weeks 3-5

- ❑ Assignments 1 and 2
- ❑ Project Proposal

- ❑ Weeks 5-11

- ❑ Design, implement, test, debug, evaluate, improve – rinse and repeat

- ❑ Continually

- ❑ Team should have regular meetings among themselves
- ❑ Meetings with teaching staff as often as needed, especially weeks 3-10

Grading

- ❑ Team formation – 5%
 - ❑ Assignment 1 – 5%
 - ❑ Project Proposal – 10%
 - ❑ Assignment 2 – 5%
 - ❑ Progress Report – 25%
 - ❑ Presentation – 10%
 - ❑ Final Report – 40%
-
- ❑ Grace days – 1 per indiv assignment, 4 per project

Learning Goals

Practical AI/ML

- Be creative about ideas
- Understand what's practical
- Implement and debug algorithms
- State-of-art AI application!

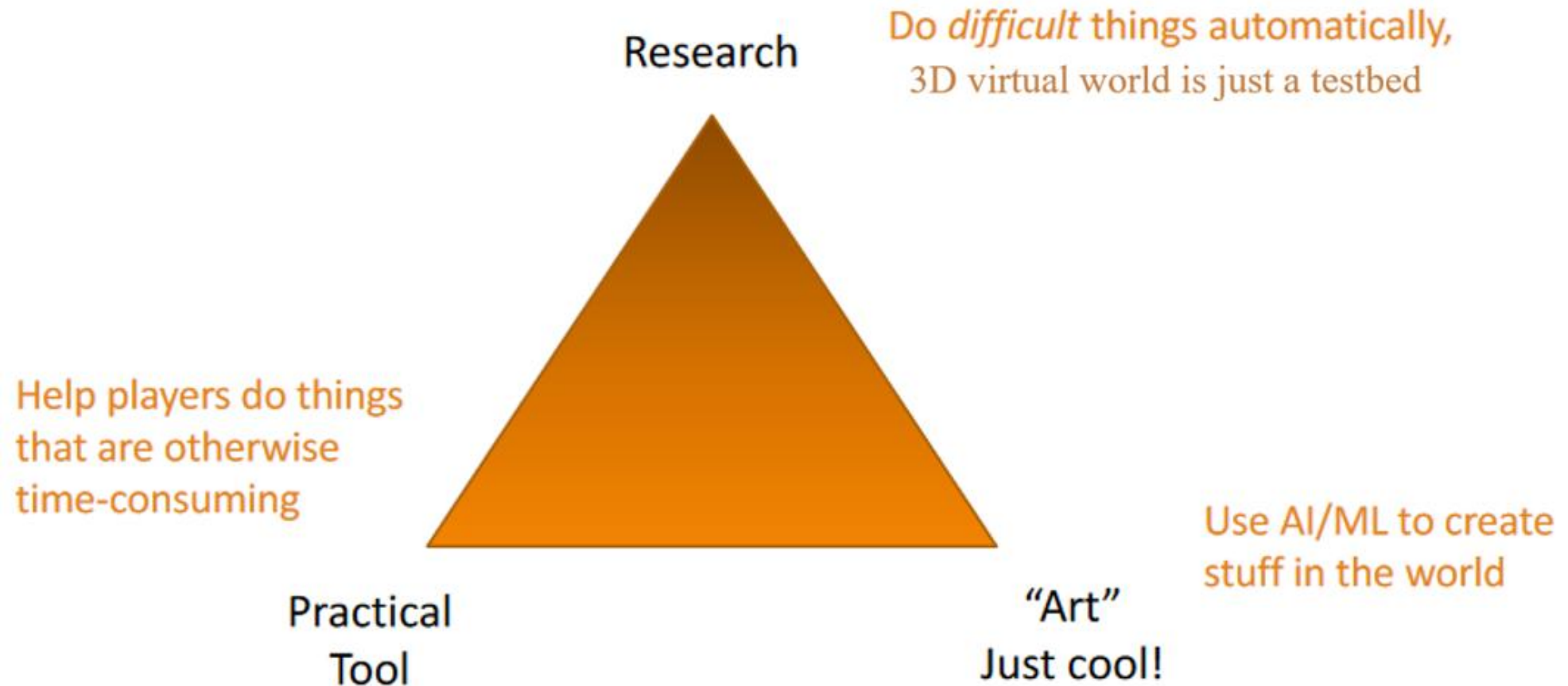
Software Engineering

- Design a complex software system
- Use modern software practices
- Learn to program collaboratively
- Small team of developers!

Presentation Skills

- Be able to “sell” your idea in writing/images/videos
- Present your project in a convincing manner
- Document and maintain a website
- Public presence of the project!

What can a project be?



Technical solution

Use Artificial Intelligence or Machine Learning algorithms

Artificial Intelligence

Heuristic/Adversarial/Local Search

Logic

Planning

Reinforcement Learning

Computer Vision

Constraint Satisfaction

Machine Learning

Supervised Learning

Bayesian Networks

Unsupervised Learning

Natural Language Processing

Recommendation Systems

Computer Vision

Time Series Modeling

Deep Learning

Evaluation

How would YOU define that your project was a success?



Quantitative
Evaluation

Numerical Metrics:

- Accuracy, F1, AUC, ...
- Time to “run”, time to “train”

Baselines:

- What would be currently used?
- What are reasonable “simpler” methods?

By how much amount?

We hope to improve the METRIC by AMOUNT over BASELINE!

(I won't hold you to it, just want you to think about it)

Evaluation

How would YOU define that your project was a success?



Qualitative
Evaluation

Simple Example Cases:

- What are examples that your idea will “definitely” work on?
- What is the expected output on these?

Error Analysis and Introspection:

- Are there plots/figures to verify the behavior?
- If it doesn’t work, how will you improve it?

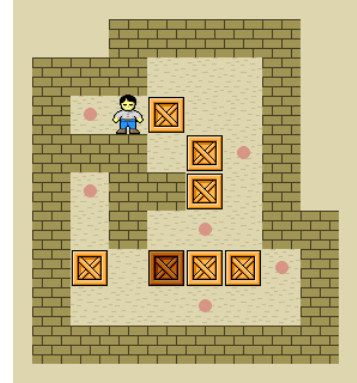
The Super-Impressive Example

- What is the best example? “awesome if it works”
- E.g. something that perfectly captures your idea!

Topics : planning/control in 3D virtual world



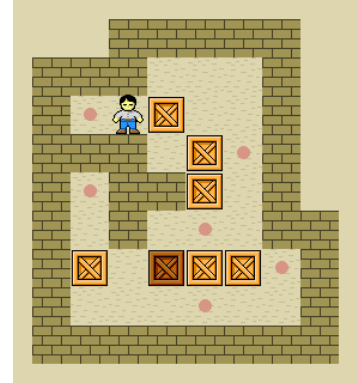
- ❑ Gymnasium (<https://github.com/Farama-Foundation/Gymnasium>) formerly OpenAI
 - ❑ Build an agent that operates in a virtual 3D world
 - ❑ Learn about the world and control agent using API
 - ❑ Pick a particular task, implement agent to carry out this task
 - ❑ Mostly Reinforcement Learning tasks
 - ❑ Driving a car & navigating (low level – turn wheels, apply brake/acceleration (camera sensor data); high level – plan trip & navigate traffic (traffic data)), chasing a target, manipulate objects, train robot, controlling a drone swarm, etc.
- ❑ Malmo – ok, more tasks, but has not been updated in a few years
 - ❑ Explore 3D world, carry out a wide variety of tasks – extract material, craft tools, build structures, fight, socialize, etc.
 - ❑ Malmo platform – full access to Minecraft information, objects, rewards, etc.



Topics : other



- ❑ Puzzles, e.g. Sokoban, Sudoku, etc. – no feedback from environment
- ❑ Turn-taking single-player games, e.g. Minesweeper, Wordle, etc. – feedback from environment
- ❑ Combinatorial optimization problems, e.g. Traveling Salesman, 3SAT, N-queens, etc.
- ❑ Query-answer tasks in formal models, e.g. Graphical Models, Bayesian Networks, etc.
- ❑ Machine Learning, Data Science, e.g. LLM-based
 - ❑ Subject to approval case-by-case
- ❑ Something not on this list
 - ❑ requires a case-by-case approval



Assignments



Assignments

- Beginning of the quarter

Objective

- Familiarize yourself with the platform/task/background
- Implement at least one algorithm

Source Code (Python)

- Submit relevant source code
 - We will not run it, but will read it
- Statement of collaboration, if any
 - Only limited discussions allowed

Meetings

Lectures

- In-person lectures for first few weeks
- After that, only work on the project

Discussions

- **TBA**
- Office hours with staff

Additional Meetings

- Expect to regularly meet other students
- There will be a few mandatory meetings with the staff

Office Hours

Instructor

- Room: Zoom only, by appointment only
- Meet to discuss project ideas, progress, roadblocks, evaluation plans, etc.

TAs

- Song, Yutong – yutons12
- TBA

Course Project



Groups for the Project

- Team size should be 3

Course Project

- ❑ Handling AI is just one part
- ❑ Also need to manage the software engineering part
 - ❑ Requirements spec
 - ❑ Code design
 - ❑ Documentation
 - ❑ Collaboration tools, code repository
 - ❑ Code reviews
 - ❑ Milestones, development targets
 - ❑ Unit testing, system testing
 - ❑ Etc.
- ❑ You will not be graded on software project management (this is not software/programming course) but good software development practices are important

Use Github

